Modeled Redistribution of Mercury Mobilized from Different Sources within Upper Galveston Bay during Hurricane Harvey

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AGU Fall Meeting, Thursday December 12, 2024, Washington DC USA. | ***Poster #OS41I-0552***

Introduction

Hurricane Harvey made landfall along the mid-Texas coast on August 26, 2017, as a Category-4 hurricane, dumping record-breaking precipitation to Galveston Bay. The freshwater load into the bay was estimated to be 11 x 10^9 m^3, equivalent to 3 times the total volume of the bay [8]. The bay experienced compound flooding, resulting from the interaction between the oceanic storm surges as the storm approached the coast, and peak flood pulses from rainfall in the upper reaches of the bay, Buffalo Bayou and San Jacinto River [3]. Based on analysis of sediment core data from before and after the hurricane, the flooding mobilized thick layers of material from the seabed, including mercury-contaminated sediment in the upper streams[1].

Enrichment factors for contaminants indicate that human activities are the primary source of mercury in the sediments of Galveston Bay[7]. Sediment cores collected in Galveston Bay prior to Hurricane Harvey suggest a high mercury (Hg) concentration in upstream areas [4]. Patrick Bayou, a small tributary in lower Buffalo Bayou and a U.S. EPA Superfund Site, has been considered the dominant source of legacy mercury [1,2].

Study Site

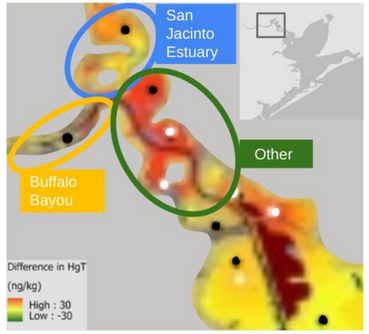

Figure 1: Galveston Bay HgT concentration difference edited from Dellapenna et al. 2020, 2022[1,2]

Galveston Bay is a natural semi-enclosed estuarine system located on the Texas Coast in the Gulf of Mexico, and it is ranked as the 7th largest estuary in the United States[1]. The bay is a micro-tidal, wind-dominated estuary with an average depth less than 3 m and a narrow tidal inlet opening to the Gulf [8]. The image in the right hand corner of figure I shows the entirety of the Bay. The results are displayed only in the model domain.

The map above, shows the difference in total mercury (HgT) concentrations on the seabed before and after Hurricane Harvey based on sediment core data. White dots represent locations sampled before the hurricane and black dots represent samples taken after the storm. These data were used to approximate changes to HgT concentration in unmeasured locations to complete the map. Buffalo Bayou and the San Jacinto Estuary were not sampled before the hurricane, so concentrations there are estimates.

Research Goal

Our goal was to fill a gap in understanding of the source-to-sink process concerning mercury mobilization and deposition. We wanted to answer the question:

Which source locations contribute to downstream deposition?

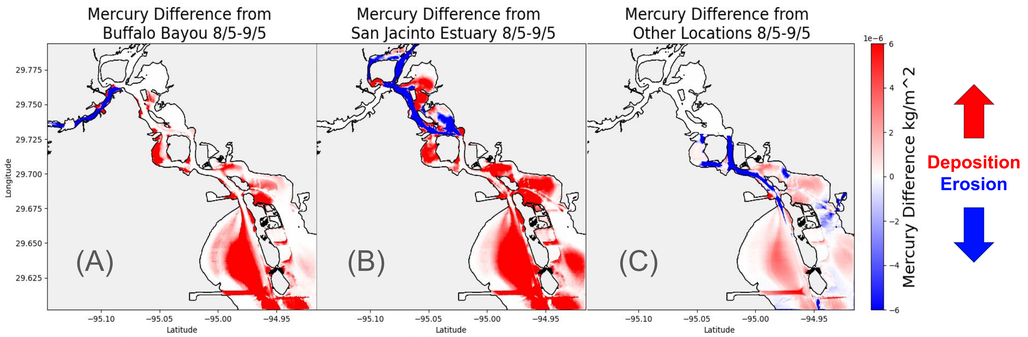

Figure 2: Difference in mercury concentration from different source locations

Methods and Model

Used the Regional Ocean Modeling System (ROMS):

- Structured 3-D hydrodynamic model
- 20m horizontal resolution
- 6 water column layers
- Accounted for transport of mercury bound to sediment

Time series inputs:

- Wind Speed from CSFV2
- River discharge from LSU WRF-hydro
- Open boundary water level from SCHISM

Three different tracers for mercury were defined representing mercury concentrations in upper San Jacinto Estuary, lower Buffalo Bayou, and Burnett Bay (Figure 1).

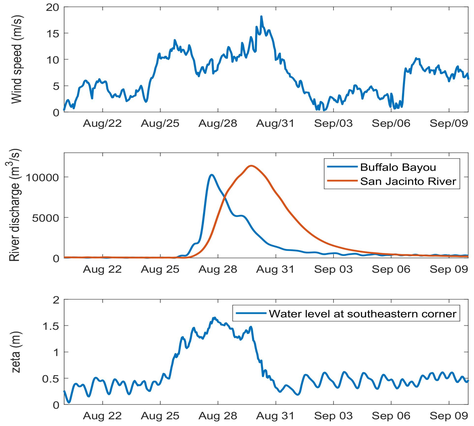

Figure 3: by Zhiyun Du 2024[3]

Results and Analysis

The redistribution of mercury sourced from the three locations are shown in the panels above:

A. Significant amounts of mercury were eroded from Buffalo Bayou and deposited in the lower model domain.
B. Mercury eroded from the upper San Jacinto Estuary was distributed across most regions within the model domain, including a major deposition within the San Jacinto Estuary.
C. Other locations containing mercury did experience erosion, but not much of this was deposited in the model domain.

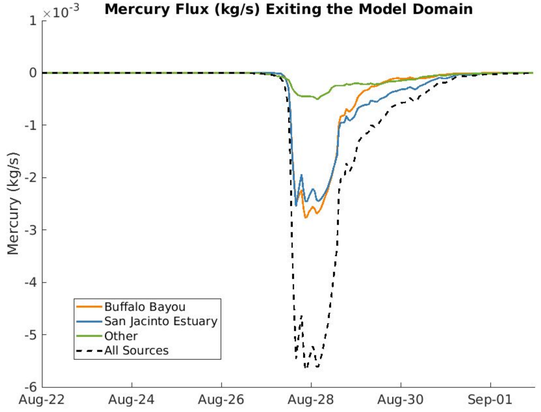

Figure 4: Time series of flux from different sources

The time series, figure four, flux calculated through the eastern and southern boundaries of the model in kilograms per seconds.

- Negative values represent mercury leaving the model domain.
- Model ran for August 2027
- Hurricane Harvey made landfall on August 25th. Preceding the storm, most mercury was retained in the model domain.
- Peak export occurred on August 28th, which coincided with the peak river discharge.
- The fluctuations observed in the export graph are believed to be caused by tidal exchanges at the boundary.

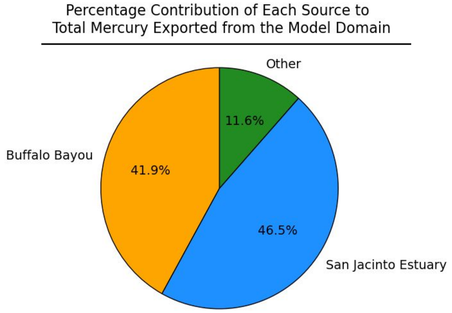

Figure 5: Pie chart of source contribution to downstream.

The pie chart shows the proportion of exported mercury that was sourced from the three areas considered. The San Jacinto Estuary contributed the most to export, though only slightly more than Buffalo Bayou. In comparison, the other locations contributed very little to export. Consequently, the model results indicated that contaminated sediment eroded from both Buffalo Bayou and the San Jacinto estuary served as sources for mercury delivered to Galveston Bay during this storm.

Future Work

Moving forward, the model's sensitivity to initial mercury concentration, particulate settling velocities, and chemical properties should be researched. A higher-resolution model of Patrick Bayou, a Superfund site, could yield valuable insights. These are steps toward a long-term goal of developing operational models for contaminant transport during storms.

References

1 Dellapenna, T. M., Hoelscher, C., Hill, L., Critides, L., Salgado (Bartlett), V., Bell, M., Al Mukaimi, M. E., Du, J., Park, K., & Knap, A. H. (2022). Hurricane Harvey delivered a massive load of mercury-rich sediment to Galveston Bay, TX, USA. *Estuaries and Coasts, 45*(3), 428–444.

2 Dellapenna, T. M., Hoelscher, C., Hill, L., Al Mukaimi, M. E., & Knap, A. (2020). How tropical cyclone flooding caused erosion and dispersal of mercury-contaminated sediment in an urban estuary: The impact of Hurricane Harvey on Buffalo Bayou and the San Jacinto Estuary, Galveston Bay, USA. *Science of The Total Environment, 748*, Article 141226.

3 Du, Z. (2024). Modelinh Mercury Transport from Upstream Reaches of Galveston Bay During Hurricane Harvey [Poster presentation]. American Geophysical Union, Washington, DC.

4 Valle-Levinson, A., Olabarrieta, M., & Heilman, L. (2020). Compound flooding in Houston-Galveston Bay during Hurricane Harvey. *Science of the Total Environment, 747*, 141272.

5 Al Mukaimi, M. E., Kaiser, K., Williams, J. R., Dellapenna, T. M., Louchouarn, P., & Santschi, P. H. (2018a). Centennial record of anthropogenic impacts in Galveston Bay: Evidence from trace metals (Hg, Pb, Ni, Zn) and lignin oxidation products. *Environmental Pollution, 237,* 887-899.

6 Al Mukaimi, M. E., Dellapenna, T. M., & Williams, J. R. (2018b). Enhanced land subsidence in Galveston Bay, Texas: Interaction between sediment accumulation rates and relative sea level rise. *Estuarine, Coastal and Shelf Science, 207,* 183-193.

7 Warner, J. C., Sherwood, C. R., Signell, R. P., Harris, C. K., & Arango, H. G. (2008). Development of a three-dimensional, regional, coupled wave, current, and sediment-transport model. *Computers & Geosciences, 34*(10), 1284-1306.

8 Lopez, B. E., Allen, J. M., Dukes, J. S., & Bradley, B. A. (2022). Global environmental changes more frequently offset than intensify detrimental effects of biological invasions. *Proceedings of the National Academy of Sciences, 119*(22), e2117389119.

9 Du, J., Park, K., Dellapenna, T. M., & Clay, J. M. (2019). Dramatic hydrodynamic and sedimentary responses in Galveston Bay and adjacent inner shelf to Hurricane Harvey. *Science of The Total Environment, 653,* 554-564.

10 U.S. Environmental Protection Agency. (n.d.). Superfund site: Coliseum Mine. Retrieved June 26, 2024

11 Vecta. (n.d.). *USA TX Galveston Bay [Vector map]*. Vecta. Retrieved December 4, 2024, from https://vecta.io/symbols/281/ecosystems-maps/137/usa-tx-galveston-bay